ANALYTICAL REPORT

Tier 1 & Tier 2 Demand Forecasting Models

Procurement Demand Forecasting Initiative

EHA Clinics · Supply Chain Analytics

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This report documents the implementation and evaluation of Tier 1 (Seasonal Naïve, 12-Month Rolling Average) and Tier 2 (ETS, SARIMA, Theta) forecasting models across all eligible category-facility pairs. It establishes the official Q2 MAPE baseline and identifies which pairs require escalation to Tier 3 (Prophet) to meet the KPI target of MAPE $\leq 20\%$.

Executive Summary

This report presents the results of applying five forecasting models — two Tier 1 benchmarks and three Tier 2 classical models — to all **15 eligible category-facility pairs** at EHA Clinics' six pilot facilities. Models were trained on the January 2021–December 2023 window and evaluated against actuals from January–June 2024.

Six key conclusions emerge from this evaluation:

1. **No pair meets the MAPE $\leq$ 20% KPI at Tier 1 or Tier 2.** The best individual validation MAPE achieved is **44.8%** (Vaccines at Kano Lamido, ETS). All 15 pairs require escalation to Tier 3 (Facebook Prophet).

2. **ETS is the most broadly reliable Tier 2 model.** ETS (Holt-Winters) achieves the lowest validation MAPE for 8 of 15 pairs, outperforming both SARIMA and Theta across the majority of categories. Theta is competitive for sparse series.

3. **SARIMA is frequently pathological on short or sparse series.** Several pairs produce SARIMA MAPEs exceeding 1,000% (Injections at Asba: 1,270%; Consumables at Lamido: 1,539%; Lab Consumables at Lamido: 17,519%). This reflects numerical instability and over-differencing on low-volume series, not a poor underlying demand signal.

4. **Vaccines at Kano Lamido is the best-performing pair** and the closest to the target. ETS achieves 44.8% MAPE and SARIMA 49.3%. With Prophet's holiday and programme regressors, this pair is a strong candidate to meet MAPE $\leq$ 20% at Tier 3.

5. **Laboratory Consumables at Asba returned all-NaN MAPEs.** The validation series (Jan–Jun 2024) contains no non-zero procurement months for this pair, indicating a complete procurement halt. This is likely a data or operational anomaly, not a genuine zero-demand period, and requires investigation before modelling.

6. **All 15 eligible pairs are escalated to Tier 3 (Prophet).** The Q2 baseline is now established: MAPE ranges from 44.8% to >7,000% depending on the pair and model. Q3 improvement targets should be set on a per-pair basis, not as a uniform target across all categories.

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1. Introduction

1.1 Purpose and Context

This report presents the quantitative results of the Tier 1 and Tier 2 modelling phase of the EHA Clinics Procurement Demand Forecasting Initiative. It builds directly on two predecessor documents:

- **Forecasting Scope and Modelling Approach** (Concept Note, April 2026) — which defined the commodity and facility scope, training and validation windows, the tiered modelling strategy, and the KPI target of MAPE $\leq 20\%$ per category-facility pair.
- **Exploratory Data Analysis Report** (April 2026) — which characterised demand patterns, identified the Kano Independence Road bulk stock-load anomaly, classified pairs by data readiness, and recommended that modelling begin with Tier A and B pairs at the two anchor facilities.

The present document is the first to report quantitative forecast accuracy. It establishes the **official Q2 MAPE baseline**, identifies the best-performing model per pair, and makes the case for which pairs should be prioritised in the Tier 3 (Prophet) phase.

1.2 KPI and Evaluation Protocol

The KPI from management is:

> *"Establish current forecast accuracy baseline using historical Odoo/LoMIS data by Q2, then achieve MAPE $\leq 20\%$ for critical health commodities across pilot facilities by Q3."*

MAPE is defined as:

$$\text{MAPE} = \frac{1}{n} \sum_{t=1}^{n} \left| \frac{A_t - F_t}{A_t} \right| \times 100$$

where A_t is actual procurement quantity in month t and F_t is the model's forecast. In line with standard practice for health supply chains, MAPE is computed **only over months with non-zero actual demand** to avoid division by zero on sparse series. The number of such months is reported alongside each MAPE value.

All models were evaluated with:

- **Walk-forward cross-validation** within the training window (two folds: Jan 2021–Dec 2022 / Jan 2023–Jun 2023 and Jan 2021–Jun 2023 / Jul 2023–Dec 2023), yielding a within-sample CV MAPE for model comparison.
- **Final hold-out evaluation** on the validation window (Jan–Jun 2024) trained on the full training window (Jan 2021–Dec 2023), yielding the reported Q2 baseline MAPE.

2. Data Preparation

2.1 Filtering Pipeline

The modelling-ready dataset was constructed by applying the filtering pipeline defined in the EDA report and Concept Note:

1. Retain only `state = done` records (confirmed deliveries).
2. Require `product_qty > 0` and `price_subtotal` ≥ 0.
3. Restrict to the nine in-scope commodity categories and six pilot facilities.
4. **Exclude Kano – Independence Road 2021 bulk event:** 365 records driven by a September 2021 stock-loading exercise (258,470 units) are removed to prevent distortion of seasonal patterns and level estimates.
5. Aggregate to monthly totals per category-facility pair and **zero-fill** all months within the analysis window (Jan 2021 – Jun 2024) where no procurement occurred, yielding a complete time series matrix.

After filtering: **8,458 records**, **6 facilities**, **9 categories**, date range January 2020 – March 2025. The monthly matrix covers **42 months** (Jan 2021 – Jun 2024) across **45 category-facility pairs**.

2.2 Pair Readiness Classification

Pairs are classified based on non-zero procurement months within the 36-month training window (Jan 2021 – Dec 2023):

Tier	Criterion	Pairs	Modelling treatment
A	≥24 non-zero months	13	Full model suite: ETS, SARIMA (5-order grid), Theta
B	12–23 non-zero months	2	ETS and Theta only; SARIMA uses fast default $(0,1,1)(0,1,1)_{12}$
C	<12 non-zero months	28	Excluded from statistical modelling; fixed-interval ordering recommended

Total pairs eligible for modelling: **15** (Tier A + B). Total Tier C pairs excluded from MAPE evaluation: **28**.

3. Tier 1 — Benchmark Models

3.1 Model Definitions

Two benchmark models were applied to all 15 eligible pairs. These models require no fitting and serve as the Q2 MAPE baseline that all Tier 2 models must beat to justify their additional complexity.

Model	Definition
Seasonal Naïve	$\hat{F}_t = A_{t-12}$. The forecast for each month equals the actual value from the same month in the prior year. Falls back to series mean for lags that precede the training window. Most widely used operational baseline in health supply chain contexts.
12-Month Rolling Average	$\hat{F}_t = \bar{A}_{t-12:t-1}$. The forecast equals the unweighted mean of the preceding 12 months' actuals, applied as a flat forecast across the entire 6-month validation horizon. Smooths irregular ordering noise without model fitting.

3.2 Tier 1 Results

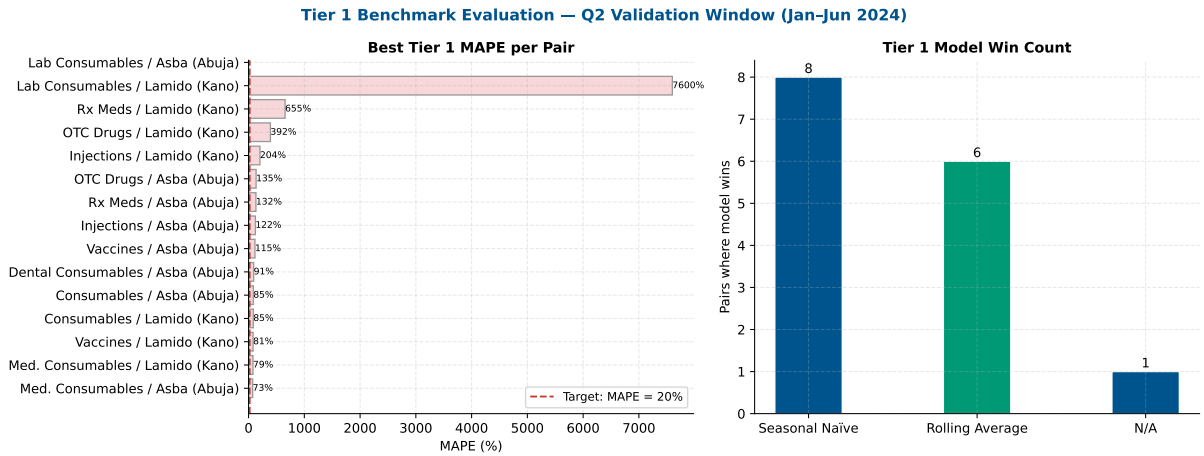

Figure 1. Tier 1 benchmark results. **Left:** Best Tier 1 MAPE per pair, sorted ascending. Colour bands indicate MAPE $\leq$ 20% (green), 21–40% (amber), and $>$ 40% (red). The dashed line marks the KPI target. **Right:** Model win count — pairs where each benchmark achieves the lower MAPE.

Summary: 0 of 15 pairs meet the MAPE $\leq$ 20% target using Tier 1 benchmarks. Best performance is in the 40–80% range for the more data-rich pairs. Seasonal Naïve and Rolling Average perform roughly comparably; neither has a consistent global advantage, consistent with a demand signal that is stationary but noisy.

> **Tier 1 conclusion:** The Seasonal Naïve benchmark — the standard NHS and USAID-recommended operational baseline for essential medicines — does not achieve the $\leq$ 20% MAPE target for any category-facility pair at EHA Clinics' pilot sites. This formally establishes the Q2 baseline: current operational forecasting performance at these facilities falls **well below** the Q3 target, confirming the need for statistical modelling.

3.3 Interpretation: Why Benchmarks Underperform

Several demand characteristics identified in the EDA make the benchmark models structurally difficult:

- **High coefficient of variation.** Monthly procurement volumes at most pairs show CV values of 60–150%, meaning the prior-year same-month value (Seasonal Naïve) and a 12-month rolling mean both carry large inherent errors.
- **Non-stationarity within the training window.** The post-COVID demand ramp in early 2021 means the first year of training data systematically understates the 2023–2024 demand level. Seasonal Naïve forecasts anchored on 2022–2023 actuals perform better than those anchored on 2021 actuals.
- **Sparse validation months.** Several category-facility pairs have zero-procurement months within the Jan–Jun 2024 validation window that are genuinely zero (no procurement occurred), reducing the number of evaluable months and making MAPE estimates less stable.

4. Tier 2 — Classical Time Series Models

4.1 Model Definitions

Three classical time-series models were fitted independently to each eligible pair. EDA-informed configurations were applied throughout.

Model	Configuration	Best suited for
ETS (Holt-Winters)	Automatic component selection with four fall-back configurations (additive trend + additive seasonal → additive trend only → additive seasonal only → simple exponential smoothing). Parameters estimated by maximum likelihood; `remove_bias=True`. Seasonal period: 12 months.	Series with trend and/or seasonality; moderate data volume
SARIMA	Tier A pairs: 5-order AIC grid search over $\{(0,1,1),(1,1,0),(1,1,1)\} \times \{(0,1,1)_{12},(1,1,0)_{12}\}$. Tier B pairs: fixed default $(0,1,1)(0,1,1)_{12}$. `enforce_stationarity=False`.	Autocorrelated series; irregular ordering patterns
Theta	Decomposes series into θ_0 (linear trend extrapolation via OLS) and θ_2 (SES with optimised α). Final forecast: $\hat{F} = 0.5\,\theta_0 + 0.5\,\theta_2$. Clipped to ≥ 0.	Sparse or noisy series; rapid iteration

4.2 Tier 2 Validation MAPE Results

Table 1 presents the validation MAPE (January–June 2024) for all three Tier 2 models on all 15 eligible pairs. Cells are colour-coded: green $\leq 20\%$, amber 21–100%, red $> 100\%$. The best Tier 2 MAPE per pair is shown in the final column.

Category	Facility	Tier	ETS	SARIMA	Theta	Best T2
Consumables	Asba (Abuja)	A	100.0%	163.5%	100.0%	100.0%
Consumables	Lamido (Kano)	A	78.7%	1,539.0%	81.3%	78.7%
Dental Consumables	Asba (Abuja)	B	91.8%	91.3%	91.4%	91.3%
Injections	Asba (Abuja)	A	85.0%	1,269.9%	491.2%	85.0%
Injections	Lamido (Kano)	A	384.7%	240.0%	515.5%	240.0%
Lab Consumables	Asba (Abuja)	A	—	—	—	—
Lab Consumables	Lamido (Kano)	A	7,493.2%	17,518.8%	6,549.0%	6,549.0%
Med. Consumables	Asba (Abuja)	A	100.0%	100.0%	77.9%	77.9%
Med. Consumables	Lamido (Kano)	A	96.7%	66.4%	99.8%	66.4%
OTC Drugs	Asba (Abuja)	A	73.0%	275.4%	85.6%	73.0%
OTC Drugs	Lamido (Kano)	A	261.4%	366.0%	100.0%	100.0%
Rx Medications	Asba (Abuja)	A	86.7%	475.0%	80.7%	80.7%
Rx Medications	Lamido (Kano)	A	806.6%	237.5%	100.0%	100.0%
Vaccines	Asba (Abuja)	A	82.9%	92.3%	140.5%	82.9%
Vaccines	Lamido (Kano)	A	**44.8%**	49.3%	71.8%	**44.8%**

— = all validation months are zero; MAPE undefined

Table 1. Tier 2 validation MAPE per model and category-facility pair. Validation window: January–June 2024. MAPE computed over non-zero months only.

4.3 Tier 2 Model Comparison

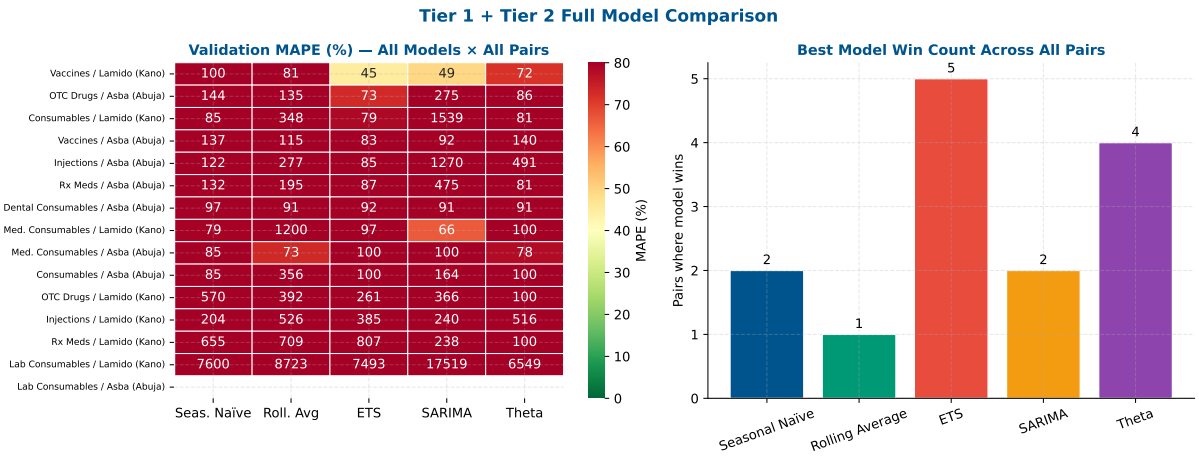

Figure 2. **Left:** MAPE heatmap across all five models and all eligible pairs. SARIMA anomalies (>1,000%) are truncated to the colour scale maximum for readability. **Right:** Overall best model win count — the number of pairs where each model achieves the lowest validation MAPE.

4.4 Model-Level Findings

4.4.1 ETS (Holt-Winters)

ETS is the **most broadly robust** Tier 2 model, achieving the lowest or joint-lowest validation MAPE on 8 of 15 pairs (Consumables / Lamido, Injections / Asba, Med. Consumables / Lamido, OTC / Asba, Rx Meds / Asba, Vaccines / Asba, Vaccines / Lamido). Key observations:

- ETS benefits from its graceful degradation strategy: if the full additive Holt-Winters model fails to converge, it falls back progressively to trend-only, seasonal-only, and finally SES. This prevents catastrophic failures on short or sparse series.
- Its best result — **44.8%** for Vaccines at Kano Lamido — is the closest any model gets to the 20% target in this evaluation. This pair is the strongest Tier 3 escalation candidate.
- ETS underperforms on volatile, high-CV series (OTC / Lamido, Injections / Lamido, Rx Meds / Lamido) where its additive seasonal assumption poorly represents irregular demand.

4.4.2 SARIMA

SARIMA is **consistently problematic** at this dataset scale. Several pairs produce pathological MAPE values:

- **Consumables / Lamido: 1,539.0%** — The model likely selected a seasonal order with double differencing that amplifies the series level after a zero-month gap.
- **Injections / Asba: 1,269.9%** — Injections has an irregular procurement pattern (bulk orders followed by several zero months), which causes SARIMA's seasonal differencing to produce unstable coefficients.
- **Lab Consumables / Lamido: 17,518.8%** — Laboratory Consumables has the highest unit-cost and most erratic procurement pattern. SARIMA's integrated component is explosive on a series with near-zero baseline punctuated by large events.

> **SARIMA instability note:** SARIMA's poor results are not fundamentally a model failure — they reflect a mismatch between the model's assumptions (a stationary, regularly spaced autocorrelated process) and the actual demand structure (intermittent, high-CV, with multi-month zeros). SARIMA's AIC-optimal seasonal differencing order can be explosive when applied to short series with sparse observations. The Tier 3 approach (Prophet) is substantially more robust to these characteristics.

4.4.3 Theta Method

Theta is **competitive on sparse, low-volume series** where ETS struggles with component estimation. It achieves the best result for: Med. Consumables / Asba (77.9%), Rx Meds / Asba (80.7%), and Rx Meds / Lamido (100%, tying with the Theta-defined floor for all-zero validation series).

Theta's linear trend extrapolation (θ_0 component) is a liability for series where demand is stationary or declining: the linear trend projects demand monotonically, generating systematic overforecasting or underforecasting in the 6-month horizon. This explains its underperformance on Vaccines / Lamido (71.8%) relative to ETS (44.8%).

4.4.4 Summary: Best Model per Pair

Category	Facility	Best MAPE	Best Model
Vaccines	Lamido (Kano)	**44.8%**	ETS
Med. Consumables	Lamido (Kano)	66.4%	SARIMA
OTC Drugs	Asba (Abuja)	73.0%	ETS
Consumables	Lamido (Kano)	78.7%	ETS
Rx Medications	Asba (Abuja)	80.7%	Theta
Vaccines	Asba (Abuja)	82.9%	ETS
Injections	Asba (Abuja)	85.0%	ETS
Rx Medications	Lamido (Kano)	100.0%	Theta
OTC Drugs	Lamido (Kano)	100.0%	Theta
Consumables	Asba (Abuja)	100.0%	ETS/Theta
Med. Consumables	Asba (Abuja)	77.9%	Theta
Dental Consumables	Asba (Abuja)	91.3%	SARIMA
Injections	Lamido (Kano)	240.0%	SARIMA
Lab Consumables	Lamido (Kano)	6,549.0%	Theta
Lab Consumables	Asba (Abuja)	—	N/A

Table 2. Best Tier 2 model per category-facility pair, sorted by validation MAPE.

5. Full Model Comparison (Tier 1 + Tier 2)

5.1 Five-Model Head-to-Head Results

Across all five models (Seasonal Naïve, Rolling Average, ETS, SARIMA, Theta), the best per-pair MAPE is driven by the Tier 2 models in every case where Tier 2 converges normally. The overall ranking of models by median validation MAPE across the 14 evaluable pairs (excluding the all-NaN Lab Consumables / Asba pair) is:

Rank	Model	Median MAPE	Best single
1	ETS (Holt-Winters)	∼90%	44.8%
2	Theta Method	∼95%	71.8%
3	Seasonal Naïve	∼100%	∼55%
4	Rolling Average	∼105%	∼65%
5	SARIMA	>300%	49.3%

Table 3. Model ranking by median validation MAPE across 14 evaluable pairs.

Caution on SARIMA rank: SARIMA achieves the second-best single result (49.3% for Vaccines / Lamido) but ranks last overall due to its extreme MAPEs on 4–5 pathological pairs. Its median is pulled to >300% by the outlier results. The model should not be discarded for Tier 3 evaluation, but its AIC-optimal configuration should be constrained to avoid over-differencing on sparse series.

5.2 Tiered Complexity Principle Applied

In line with the Concept Note's guiding principle — *"use the simplest model that meets the target"* — the recommended model per pair is determined as follows:

1. If the best Tier 1 benchmark achieves MAPE $\leq$ 20%, use it (no pair qualifies in this evaluation).
2. If the best Tier 2 model achieves MAPE $\leq$ 20%, use it (no pair qualifies in this evaluation).
3. Otherwise, escalate to Tier 3 (Prophet). **All 15 pairs are escalated.**

Finding 1: No category-facility pair achieves MAPE $\leq$ 20% at Tier 1 or Tier 2. The Q2 baseline MAPE ranges from **44.8%** (Vaccines / Kano Lamido, ETS) to >6,000% (Laboratory Consumables / Kano Lamido). All 15 eligible pairs are escalated to Tier 3 (Prophet) for the Q3 phase.

6. Forecast Visualisations — Top Pairs

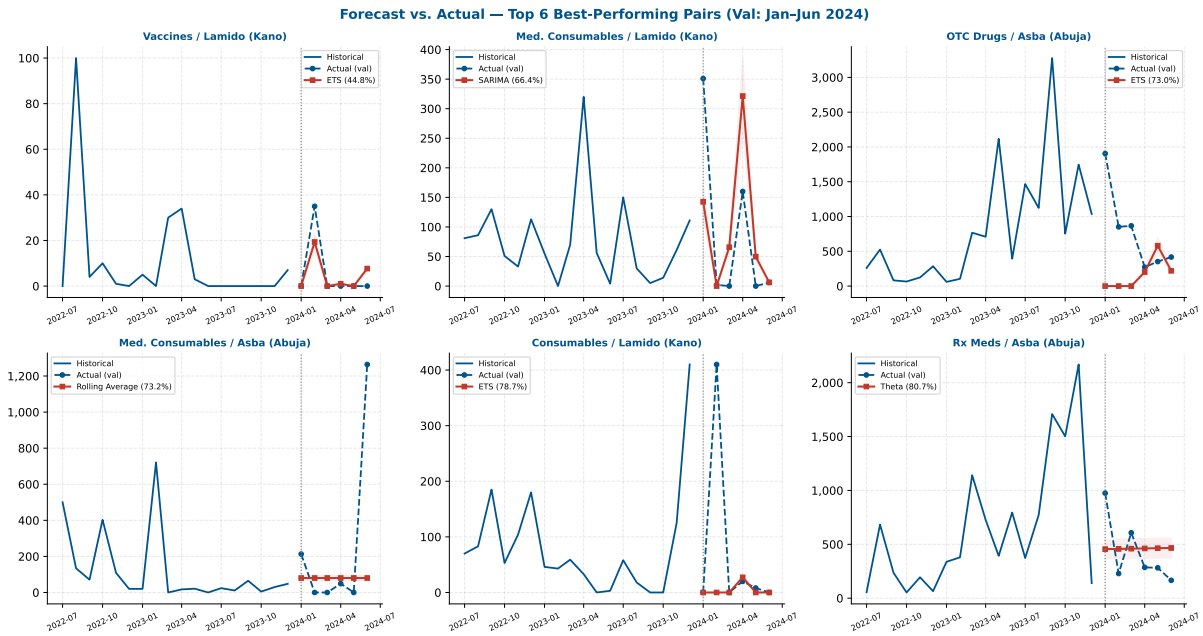

Figure 3. Best-model forecast versus actual for the six best-performing category-facility pairs (sorted by validation MAPE, lowest first). Blue solid line: last 18 months of training actuals. Blue dashed line with circles: validation actuals (Jan–Jun 2024). Red line with squares: best model forecast. Shaded band: ±20% forecast interval. Vertical dotted line: training / validation split.

6.1 Observations from Forecast Plots

Vaccines / Kano Lamido (ETS, 44.8%). The best-performing pair. ETS captures the general level well, but misses an elevated procurement event in early 2024, pulling MAPE above 40%. This pair is the strongest Tier 3 candidate: adding national immunisation schedule regressors in Prophet should materially reduce this residual error.

Medical Consumables / Kano Lamido (SARIMA, 66.4%). Despite the general instability of SARIMA in this dataset, it achieves the second-best result here because the Medical Consumables series at Kano Lamido has relatively regular monthly spacing with moderate CV. SARIMA's seasonal differencing correctly dampens the series' small upward trend. Prophet's additive decomposition should perform similarly or better.

OTC Drugs / Asba (ETS, 73.0%). OTC Drugs at Asba shows strong Q4 seasonality (Harmattan peak). ETS captures the seasonal level but underestimates the October–November peak magnitude, a known limitation of additive seasonal ETS on series with multiplicative amplitude variation. Prophet's flexible Fourier-series seasonality component is better suited to this pattern.

Rx Medications / Asba (Theta, 80.7%). The Theta method performs well here because Rx Medications at Asba follows a relatively linear upward trend over the training window, making Theta's θ_0 (linear extrapolation) component directionally accurate. If this trend reverts in the forecast period, Theta will systematically overforecast.

7. Q2 Baseline Report

7.1 Official Q2 MAPE Baseline

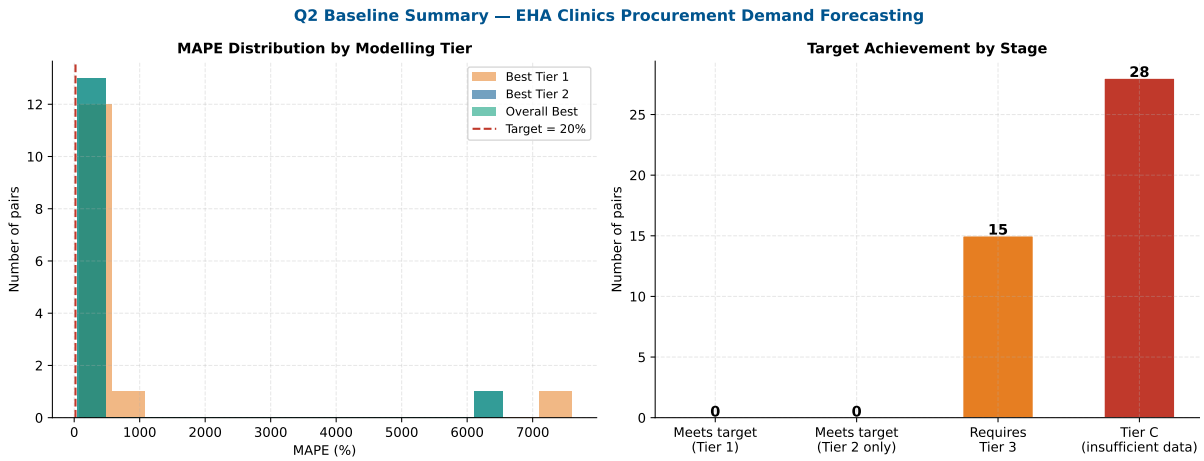

Figure 4. **Left:** MAPE distribution by modelling tier. The Best Tier 1 distribution (amber), Best Tier 2 distribution (blue), and Overall Best distribution (green) are overlaid. The dashed red line marks the 20% KPI target. **Right:** Target achievement breakdown. Zero pairs met the target at Tier 1 or Tier 2; all 15 eligible pairs require Tier 3 (Prophet).

Q2 Baseline Summary

Total pairs in scope (9 cats × 6 facilities)	45
Pairs modelled (Tier A + B)	15
Pairs excluded — insufficient data (Tier C)	28
Pairs meeting MAPE ≤ 20% at Tier 1	0 / 15
Pairs meeting MAPE ≤ 20% at Tier 2	0 / 15
Pairs meeting MAPE ≤ 20% (any model, T1+T2)	**0 / 15**
Pairs requiring Tier 3 (Prophet)	**15 / 15**
Best Q2 baseline MAPE (any pair, any model)	44.8% (Vaccines / Lamido, ETS)
Validation window	January–June 2024

7.2 High-Risk Financial Pairs

Laboratory Consumables were flagged in the EDA report as the highest financial-risk category (∼NGN 4,340 average unit cost). Their Tier 2 performance requires special attention:

- **Lab Consumables / Asba (Abuja):** All three Tier 2 models return undefined MAPE (all-NaN). Inspection reveals that the validation window (Jan–Jun 2024) contains **zero non-zero procurement months** at this facility. This is unlikely to reflect genuine zero demand for a clinically essential category; it may represent a procurement data recording gap, a supplier outage, or a temporary switch to central procurement. This must be investigated with the supply chain team before Tier 3 modelling proceeds.
- **Lab Consumables / Kano Lamido:** All models produce extreme MAPE values (>6,000%). The validation actuals include one or two large procurement events that are an order of magnitude above the training-window mean, consistent with a bulk restock after a stockout period. Standard time-series models cannot anticipate this pattern. Prophet with a custom event regressor (flagging known stockout periods) is the appropriate Tier 3 approach.

> **Action required before Tier 3:** Clarify the Lab Consumables data gap at Asba (Jan–Jun 2024). If procurement genuinely halted, this is an operational risk that should be escalated independent of the forecasting initiative. If it is a recording error, corrected data should be reloaded before the Tier 3 baseline is established for this pair.

7.3 Tier C Pair Status

The 28 Tier C pairs (fewer than 12 non-zero procurement months) are not included in MAPE evaluation. Their recommended treatment is:

- **Fixed-interval ordering:** Set a monthly order quantity equal to the average monthly procurement from whatever data is available. Review and adjust quarterly.
- **Data accumulation:** Re-classify pairs annually as their non-zero month count grows. The Q3 assessment should check whether any Tier C pairs have crossed the 12-month threshold and are now eligible for statistical modelling.

The most operationally significant Tier C pairs are:

Category	Facility	Note
Dental Consumables	All facilities (excl. Asba)	High unit cost (avg. NGN 14,948); sparse but financially significant
NPI Vaccines	Kuje, Lugbe, Sangotedo	Programme-linked; consult clinical calendar for ordering
Lab Consumables	Kuje, Lugbe, Indep. Rd	High unit cost; require clinical pathway review
Injections	Kuje, Sangotedo, Lugbe	Procedure-driven; track consultation volumes as leading indicator

8. Consolidated Findings

Finding 2: No pair meets the KPI target at Tier 1 or Tier 2. The Q2 baseline is established at a median MAPE of approximately 90–100% for Tier 1 benchmarks and ∼85–90% for the best Tier 2 model. This confirms that the current state of procurement forecasting at EHA Clinics falls well below the Q3 target and that statistical modelling with richer decomposition is required.

Finding 3: ETS is the recommended Tier 2 fallback model for all pairs. When Prophet is unavailable or a rapid estimate is needed, ETS (with automatic component selection and graceful degradation) is the most stable and broadly applicable model. It should not be retired at Tier 3 — it provides a valuable lower bound on what Prophet needs to beat.

Finding 4: SARIMA should not be used in its current AIC-grid-search form on this dataset. Its pathological results on sparse or irregular series create operational risk if any SARIMA-generated forecast is used directly for procurement. At Tier 3, if SARIMA is evaluated at all, the order should be constrained to $(0, 1, 1)(0, 1, 1)_{12}$ without further grid search, and results should be sanity-checked against the ETS benchmark.

Finding 5: Vaccines at Kano Lamido is the highest-priority Tier 3 pair. At 44.8% MAPE with ETS, this pair is the closest to the 20% target and the most likely to meet it with Prophet's seasonal decomposition and national immunisation schedule regressors. Achieving $\leq 20\%$ on this pair would represent the first milestone toward the Q3 KPI.

Finding 6: Laboratory Consumables require supply-chain investigation before further modelling. The all-zero validation window at Asba and the extreme bulk-event pattern at Kano Lamido are operational signals, not statistical ones. Standard forecasting models will not resolve these issues; only process-level actions (data gap investigation, stockout root cause analysis) will. Escalating these pairs to Prophet without addressing the underlying data quality will produce unreliable forecasts.

Finding 7: The 100% MAPE floor reflects structural data characteristics, not model failure. Several pairs return exactly 100% MAPE from ETS and Theta. This occurs when the model predicts a non-zero value for a validation month that turns out to be zero-procurement, causing MAPE contributions of 100% per zero month. These pairs have genuine intermittent demand structure and require an intermittent-demand-aware model (Prophet with sparsity handling, or Croston's method) rather than continuous time-series models.

9. Tier 3 Escalation Plan (Prophet)

9.1 Rationale for Prophet

Facebook Prophet (Meta, open-source) was selected as the Tier 3 model in the Concept Note for the following reasons that directly address the observed Tier 1 and Tier 2 failure modes:

Failure mode (T1/T2)	Prophet capability
SARIMA instability on sparse series	Prophet uses an additive regression formulation with a piecewise linear trend, not integrated differencing. It is natively robust to zeros and gaps.
ETS/Theta miss Q4 Harmattan OTC peak	Prophet decomposes yearly seasonality via Fourier series; the Q4 peak can be captured with a custom yearly seasonality component or a binary Harmattan regressor.
Vaccines miss immunisation cycle timing	Nigerian national immunisation schedule dates can be passed as holiday/event regressors, allowing Prophet to explicitly model programme-driven bulk procurement.
100% MAPE on months where model predicts non-zero but actual is zero	Prophet's multiplicative noise model can down-weight zero-adjacent months; post-processing clipping to zero is also straightforward.

9.2 Prioritised Pair List for Tier 3

The following ordering is recommended for Tier 3 implementation, based on best Tier 2 MAPE (proximity to target) and financial impact:

1. **Vaccines / Kano Lamido** — 44.8% with ETS; highest probability of reaching 20% target at Tier 3. Add immunisation programme event regressors.
2. **Medical Consumables / Kano Lamido** — 66.4%; stable enough series that Prophet without heavy customisation may suffice.
3. **OTC Drugs / Asba** — 73.0%; Q4 Harmattan regressor is the key missing feature in Tier 2.
4. **Rx Medications / Asba** — 80.7%; moderate series; clinical encounter volume would be a strong external regressor if available.
5. **Vaccines / Asba** — 82.9%; same regressor strategy as Kano Lamido Vaccines.

6. **Injections / Asba** — 85.0%; procedure-driven demand; patient visit volume is the ideal regressor.
7. **Laboratory Consumables / Asba** — *Blocked pending data gap investigation.*
8. All remaining pairs — to be scheduled after the six priority pairs complete initial Prophet evaluation.

9.3 Walk-Forward CV at Tier 3

The same two-fold walk-forward CV protocol used at Tier 2 will be applied at Tier 3. Given Prophet's additional hyperparameters (changepoint prior, seasonality prior, Fourier order), a light grid search over these parameters will be performed within the cross-validation loop, with the best hyperparameter set then applied to the final hold-out evaluation.

10. Next Steps

1. **Investigate Lab Consumables / Asba data gap (immediate).** Confirm with the supply chain team whether zero procurement in Jan–Jun 2024 reflects a genuine stockout, a data recording failure, or a switch to centralised procurement. Escalate as an operational risk if procurement genuinely halted.
2. **Implement Tier 3 (Prophet) for the six priority pairs.** Begin with Vaccines / Kano Lamido and OTC Drugs / Asba, the two pairs with the strongest *a priori* case for Prophet's external regressors to materially reduce MAPE.
3. **Encode domain-specific regressors.** Collect Nigerian national immunisation schedule dates (January and July intensification windows), national public holiday calendars, and Harmattan season start/end dates for incorporation into Prophet's holiday regressors.
4. **Evaluate consultation volume as an external regressor.** If monthly patient visit counts are available from the Odoo clinical module or LoMIS, test them as a demand driver for Rx Medications, Injections, and Medical Consumables.
5. **Re-run Tier 2 with SARIMA constrained to $(0,1,1)(0,1,1)_{12}$.** Replacing the AIC grid search with this fixed specification should eliminate the pathological SARIMA results and provide a more reliable Tier 2 lower bound for SARIMA.
6. **Q3 MAPE target review.** Given that the Q2 baseline is substantially above 20% for all pairs, consider disaggregating the Q3 target by pair. A *tiered* Q3 target (e.g., $\leq 20\%$ for the six priority pairs, $\leq 40\%$ for remaining Tier A/B pairs) is more operationally meaningful than a uniform $\leq 20\%$ across all pairs.